HYPERBOLIC GROUPS

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§0. Introduction

0.1 Let us start with three equivalent definitions of hyperbolic groups. First observe that for every finitely presented group Γ there exists a smooth bounded (i.e. bounded by a smooth hypersurface) connected domain $V \subset \mathbb{R}^n$ for every $n \geq 5$, such that the fundamental group $\pi_1(V)$ is isomorphic to Γ. A standard example of such a V is obtained as follows. Fix a finite presentation of Γ and let P be the 2-dimensional cell complex whose 1-cells correspond in the usual way to the generators and the 2-cells to the relations in Γ, such that $\pi_1(P) = \Gamma$. Then embed P into $\mathbb{R}^5$ and take a regular neighborhood of $P \subset \mathbb{R}^5$ for V.

(A) First definition: Call Γ *word hyperbolic* if there exists a constant $C > 0$ (depending on Γ and V), such that every smooth simple closed curve S in V (i.e. an embedded circle) which is *contractible* in V bounds a smooth embedded disk $D \subset V$, satisfying the following *isoperimetric inequality* (compare [Gr_1])

$$(Is_2) \qquad \text{Area } D \leq C \text{ length } S.$$

It is quite easy to see that this hyperbolicity does not depend upon the choice of V (but of course, the implied constant C does depend on V). One could also formulate (Is_2) in a combinatorial fashion (see 2.3) by using (simplically mapped) curves and disks in P.

(B) Second definition: Let Γ be a finitely generated group with a

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fixed finite system of generators and let $|\gamma|$ denote the length of a shortest word (in these generators) representing given $\gamma \in \Gamma$. Set (compare 1.1)

$$(\alpha\cdot\beta) = 1/2(|\alpha| + |\beta| - |\alpha^{-1}\beta|)$$

and call Γ word hyperbolic if there exists a constant $\delta \geqslant 0$ (depending on a choice of generators), such that every three elements α, β and γ in Γ satisfy the following *δ-version of the ultrametric triangle inequality*

$$(*) \qquad (\alpha\cdot\beta) \geqslant \min((\alpha\cdot\gamma),(\beta\cdot\gamma)) - \delta.$$

(This is equivalent to Rips' definition of hyperbolicity as well as to the definition in [Gr_5].)

It is not apriori obvious why this hyperbolicity is independent of the choice of generators. But after a closer look at (Is_2) and (*) one recognizes both as well known properties of *manifold of negative curvature*. Then it becomes clear that $Is_2 \Longleftrightarrow (*)$, though a direct proof of $Is_2 \Rightarrow (*)$ appears cumbersome. We give a rather short proof of that in 6.8 by using the *Riemann mapping theorem* and a version of *Ahlfors-Schwartz lemma*. To see how these analytic tools may help to solve a combinatorial problem we look at the following

(C) **Example**: Consider two triangulations of $\mathbb{R}^2$, say P_1 and P_2, satisfying the following two conditions.

(a) P_1 and P_2 are *uniformly locally finite*. That is there is an integer N such that every vertex in P_1 has at most N adjacent simplices and the same is true for P_2.

(b) P_1 and P_2 are *hyperbolic*. That is each vertex in P_1 (as well as in P_2) has at least seven adjacent triangles.

Then we claim the existence of simplical subdivisions P_1' and

P_2'. such that

(i) The triangulations P_1' and P_2' are isomorphic (as simplical complexes).

(ii) Each simplex in P_1 contains at most M simplices of P_1' for some integer M depending on N, and the same holds for P_2 and P_2'.

Sketch of the Proof: There exist (by an easy argument) Riemannian metrics g_1 and g_2 on $\mathbb{R}^2$ with curvatures between -1 and -1 - K for some $K \geq 0$ depending on N, such that the edges of P_1 are g_1-geodesic of unit g_1-length and those in P_2 are g_2-geodesics of unit g_2-length. Then Riemann's theorem provides a *conformal* diffeomorphism $f: (\mathbb{R}^2, g_1) \longrightarrow (\mathbb{R}^2, g_2)$. The norms of the differentials of f and f^{-1} are bounded according to Schwartz lemma,

$$\|D_f\| + \|D_{f^{-1}}\| \leq (2 + \sqrt{K})^2,$$

which allows a simplical approximation of f by the required simplical isomorphism $P_1' \longrightarrow P_2'$.

(D) **Third definition**: Let Γ be a group with a fixed finite system of generators. A metric space T is called a *tangent subcone of* Γ *at infinity* if it satisfies the following two properties.

(i) Geodesic property: For every two points t_1 and t_2 in T there exists a *geodesic segment* in T between t_1 and t_2. That is, by definition, an *isometric* map $s: [0,\tau] \longrightarrow T$ of the segment $[0,\tau] \subset \mathbb{R}$ with $\tau = \mathrm{dist}(t_1,t_2)$, such that $s(0) = t_1$ and $s(\tau) = t_2$.

(ii) (Ultra)limit property: For every finite subset $\{t_1,\dots,t_n\} \subset T$ there exists an infinite sequence of positive numbers $\epsilon_k \longrightarrow 0$ for $k \longrightarrow \infty$ and n sequences $\gamma_{ik} \in \Gamma$ for $i = 1,\dots,n$, such that

$$\epsilon_k \, |\gamma_{ik}^{-1}\gamma_{jk}| \longrightarrow \mathrm{dist}(t_i,t_j)$$

for $k \longrightarrow \infty$ and $1 \leqslant i,j \leqslant n$.

Now, call Γ word hyperbolic if every tangent subcone T of Γ is a *tree*. That is for every two distinct points t_1 and t_2 in T there exists one and only one *topological segment* $S \subset T$ between t_1 and t_2. (S is homeomorphic to $[0,1]$, such that t_1 and t_2 correspond to the ends of $[0,1]$.)

This definition is most intuitive and, possibly, best suited for generalizations. It says, in effect, that Γ with a word metric looks as a tree when observed from infinity. Notice that inequality (*) immediately implies the tree-likeness of Γ at infinity and the converse (in a more precise quantitative form) is proven in Section 6.

0.2 Examples of hyperbolic groups:

(A) It is obvious (with either of the three definitions) that free groups are word hyperbolic. It is also well known that (small cancellation) 1/6-groups satisfy Is_2 (see 4.7) and, hence, they are word hyperbolic. To get a rough idea of by how much hyperbolicity generalizes small cancellation, we look at these two classes of groups from a probabilistic point of view. Namely, we fix a generating set $\{\gamma_1,\dots,\gamma_p\}$ for $p \geqslant 2$ and then consider the set of presentations given by q relations which are cyclically irreducible words in γ_i, such that the lengths of these words lie in some interval $[\ell_1,\ell_2]$. The number $N = N(p,q,\ell_1,\ell_2)$ of such presentations is approximately $q(2p-1)^{\ell_2}$.

Fix q and let ℓ_1 and ℓ_2 go to infinity, such that the ratio ℓ_2/ℓ_1 is kept bounded, say $\ell_1 \leqslant \ell_2 \leqslant 2\ell_1$. Then, asymptotically, almost all among our presentations are 1/6. Namely, the number $N_{1/6}$ of these satisfies $N_{1/6}/N \longrightarrow 1$ as $\ell_2 \longrightarrow \infty$. However, if we drop the bound $\ell_2/\ell_1 \leqslant$ const, we come up with the opposite conclusion. That is $N_{1/6}/N \longrightarrow 0$ for $\ell_2 \longrightarrow \infty$, provided there are words in our presentation whose lengths, say ℓ_1 and ℓ_2, satisfy $\ell_2 \geqslant \exp 10p\ell_1$. Yet the number N_h of the word hyperbolic (groups given by these) presentations is still okay, $N_h/N \longrightarrow 1$.

This "genericity" of word hyperbolic groups is important in various constructions of "exotic" finitely generated infinitely presented groups (notice that word hyperbolic groups are finitely presented, see 2.2) such as *infinite torsion groups* and *Olshanski groups* which have no proper non-cyclic subgroups. These are usually obtained by taking an infinite sequence of relations w_i in fixed generators $\gamma_1,\dots,\gamma_p$ with a desired (e.g. torsion) property, and the problem is to show that the resulting group $\{\gamma_1,\dots,\gamma_p \mid w_1, w_2 \dots\}$ is infinite. The small cancellation approach becomes increasingly complicated here (for instance, see [Ol]), since intermediate finitely presented groups $\{\gamma_1,\dots,\gamma_p \mid \gamma_1,\dots,\gamma_k\}$ are not small cancellation for large $k \longrightarrow \infty$. Yet they can be chosen hyperbolic and then geometric techniques apply. In fact, once basic geometric features of negatively curved manifolds (such as the above inequalities Is_2 and (*) as well as the equivalence $Is_2 \Longleftrightarrow$ (*)) are observed (and proved) for word hyperbolic groups, then there remains nothing mysterious about "exotic" groups. They are seen as clearly as, for example, (the freedom of) subgroups in a free group, once free groups are looked upon as fundamental groups of 1-dimensional polyhedra.

For example, it is geometrically obvious (and a complete proof takes hardly a page, see 4.5.C) that the fundamental group Γ of every closed manifold of negative curvature admits an infinite factor group $\bar{\Gamma}$ all of whose elements are torsion (probably, one can achieve a universal bound on the order of the torsion, but here geometry still lags behind algebra) and such that $\bar{\Gamma}$ isometrically acts on a certain space of non-positive curvature.

In fact, one can build "exotic" groups $\bar{\Gamma}$ by adding (infinitely many) relations to any given hyperbolic group Γ (instead of the free group generated by $\gamma_1,\dots,\gamma_p$). If one starts with a lattice Γ in $Sp(n,1)$ (see (B) below) with $n \geqslant 2$, then one comes up with a $\bar{\Gamma}$ which among other "pathologies" enjoys *Kazdan's T-property* that is a sharpening of non-amenability (see 5.6).

(B) The hyperbolic groups Γ obtained by adding random relations to a free group are, after all, quite simple *two-dimensional* creatures as they (by an easy argument) act (freely if there is no torsion) on locally finite 2-dimensional polyhedra X with compact quotients $V = X/\Gamma$. One knows in many cases that the quotient polyhedron (or *orbihedron*, if Γ has torsion) V admits a (singular) metric of negative curvature (see Section 4). Similar objects exist in all dimensions $n \geqslant 2$ (see Section 4) and the most studied among them are closed Riemannian *manifolds* (and orbifolds) of negative curvature. The very existence of such manifolds V and of their (hyperbolic!) fundamental groups Γ relies (in known examples for large n, say for $n \geqslant 20$) on (elementary but non-trivial) arithmetic of number fields. Namely, these Γ are cocompact arithmetic subgroups (or simple modifications of arithmetic subgroups) in simple Lie groups with $\mathbb{R}$-rank = 1. These are $O(n,1)$, $U(n,1)$, $Sp(n,1)$ and the Cayley group. The study of discrete (not necessarily cocompact) subgroups Γ in semisimple (of any rank) Lie groups in the last thirty years revealed an astounding geometric (as well as algebraic) beauty and stimulated a search for more general classes of similar groups Γ. It seems word hyperbolic groups provide a good start for this search.

(C) Higher dimensional hyperbolic groups also can be constructed by combinatorial (or geometric) means. For example, *for every finite polyhedron* V_0, *one can find an aspherical polyhedron* V *with word hyperbolic fundamental group* $\Gamma = \pi_1(V)$, *such that* $\dim V = n = \dim V$ *and* V *admit a continuous map* $f\colon V \longrightarrow V_0$ *which is injective on the cohomology* $H^*(V_0)$. *Furthermore, if* V_0 *is a manifold, then one also can choose* V *a manifold, such that* $f^*\colon H^*(V_0;\mathbb{Q}) \longrightarrow H^*(V,\mathbb{Q})$ *sends the Pontryagin classes of* V_0 *to those of* V. (See 3.4.)

(D) Isometry groups of manifolds (and singular spaces) with non-positive curvature (e.g. lattices in semisimple Lie groups with $\mathbb{R}$-rank $\geqslant 2$) usually are not word hyperbolic. Since we do not know what is the correct definition for such groups (and we do not know if several natural definitions are equivalent) we suggest the following

(E) <u>**Non-definition**</u>: A group Γ is called *semihyperbolic* if it looks as if it admits a discrete cocompact isometric action on a space of non-positive curvature. (Sometimes we allow *non-cocompact* actions which brings into the picture finitely generated nilpotent groups and related classes of groups.)

Additional motivating examples are 1/5-groups and Cartesian products of hyperbolic (and semihyperbolic) groups. (Notice that $\Gamma_1 \times \Gamma_2$ is not word hyperbolic unless one of the two factors, Γ_1 or Γ_2, is a finite group.)

(F) <u>**Relative hyperbolicity**</u>: One can handle some weak forms of *semihyperbolicity* in a purely hyperbolic context with a notion of a group Γ which is (word) *hyperbolic relative to a given system of subgroups* in Γ. This notion (defined in 8.5) is similar to that of a small cancellation group over a free product with amalgamations. Basic examples of relatively hyperbolic groups are *non-cocompact* lattices Γ in Lie groups of $\mathbb{R}$-rank = 1. Such a Γ is hyperbolic relative to the *cuspidal* subgroups $\Gamma_i \subset \Gamma$ (which are certain virtually nilpotent groups).

In fact we give in Section 2 a definition of (non-word) hyperbolic groups which automatically includes the relative hyperbolicity (as well as discrete infinite covolume subgroups in Lie groups of $\mathbb{R}$-rank = 1).

0.3 <u>Basic features of hyperbolic groups</u>:

(A) The tree-like behaviour of hyperbolic groups Γ at infinity (see 0.1(D)) shows a strong resemblance between word hyperbolic groups and free groups. A great deal of the small cancellation theory is devoted to generalizing basic properties of free groups to small cancellation groups and a similar generalization carries over to hyperbolic groups. In fact, standard small cancellation arguments become significantly shorter when translated to the hyperbolic language. For example, the solvability of the word and conjugacy problems become totally obvious in this language.

(B) The next stage of a study of Γ starts with a construction of an *ideal boundary* (see 1.8), also called *the hyperbolic boundary* $\partial\Gamma$. This is a functorially constructed compact space where Γ acts by homeomorphisms. (This construction is essentially due to Mostow and Margulis.) For example, if Γ is a free group with $p \geqslant 2$ generators then $\partial\Gamma$ is the Cantor set formed by infinite irreducible words in these generators. This $\partial\Gamma$ is 1-dimensional for small cancellation groups (e.g., $\partial\Gamma$ is the usual boundary circle for surface groups Γ). If Γ is the fundamental group of a closed n-dimensional manifold of negative curvature, then $\partial\Gamma$ is homeomorphic to S^{n-1}, though it does not carry, in general, any natural smooth structure. The topology of $\partial\Gamma$ for a typical Γ looks quite complicated, both locally and globally, but no systematic study has been conducted so far.

(B') Using the action of Γ on $\partial\Gamma$ we construct *a compact space (of geodesics)* with an action of $\mathbb{R}$ on this space called *the geodesic flow*. This flow is not unique. Yet the one-dimensional foliation into orbits is uniquely (though not functorially) determined by Γ. This rigidity (or stability) of geodesics in manifolds of negative curvature is a famous theorem by M. Morse (see [Mor]) which was brought into the modern context of *hyperbolic dynamics* by Smale and Anosov. In fact, the geodesic flow of Γ is (Anosov-Bowen) hyperbolic. This allows us to apply to Γ the major geometric tool of hyperbolic geometry. That is Thurston's method of *geodesic (hyperbolic) simplices*. A standard application of this method shows, for example, that every *word hyperbolic group Γ contains at most finitely many pairwise non-conjugate subgroups isomorphic to a given finitely presented group Γ_0, provided Γ_0 has only one end.* (Compare 5.3.)

(B") An important feature of the hyperbolic dynamics (also discovered by Morse) is *Markov coding* (or presentation) of hyperbolic systems which bring in combinatorial techniques of *symbolic dynamics* (see [Bow]). The efficiency of the combinatorial methods is seen, for example, in Manning's proof (see [Man]) of Smale's conjecture on *the rationality* of the *ζ-function* (which counts periodic orbits) of every Bowen-hyperbolic $\mathbb{Z}$-action and Cannon's proof (see [Can]) of the rationality of the *counting function* $Z(t) = \sum_k \#\{\gamma \in \Gamma \mid |\gamma| = k\} t^k$ for fundamental groups Γ of manifolds of negative curvature. We shall see in 5.2 and 8.4 that the Markov coding applies to all hyperbolic groups and yields rationality theorems similar to those of Manning and Cannon.

(C) Every lattice Γ in a simple Lie group of rank one naturally acts on $S^{n-1} = \partial\Gamma$ and preserves certain conformal (and, hence, quasiconformal) structure (see [P]) determined by the underlying Lie group. (If $\Gamma \subset O(n,1)$, this is the usual conformal structure.) Margulis (see [Marg]) pointed out that (the first version of) Mostow's proof of rigidity for these Γ produces this conformal structure intrinsically in terms of ("algebraic" properties of) Γ. This remark by Margulis suggests the existence of a natural geometric structure for "most" word hyperbolic groups Γ. (Exceptional Γ, probably, are those where the boundary $\partial\Gamma$ is disconnected or contains a connected open subset $U \subset \partial\Gamma$, such that $U \setminus A$ is disconnected for some zero-dimensional $A \subset U$.) One wishes to extend Γ to the full automorphism group $L \supset \Gamma$ of this structure, such that L/Γ is compact, and such that L is the unique (in an obvious sense ruling out extensions like $\Gamma \times G$ for compact G) maximal cocompact extension of Γ. For majority of Γ this L is, probably, discrete, for the next class of Γ the group L is totally disconnected, and for the remaining Γ the group L is Lie. Here is a specific conjecture where such structure looks indispensable: Every torsionless word hyperbolic group Γ admits at most finitely many torsionless extensions $\Gamma' \supset \Gamma$ with finite quotients Γ'/Γ.

0.4 The idea of hyperbolicity has been lingering in combinatorial group theory since the basic work by Dehn. An extensive study of a class of word hyperbolic groups Γ with $\dim \partial\Gamma = 1$ (in the combinatorial disguise) was conducted by Olshanski (see [Ol]). Deep algebraic results on general hyperbolic groups are contained in the as yet unpublished work by I. Rips who calls them *groups with negative curvature*.

Our approach is motivated by hyperbolic phenomena in the geometry of manifolds and spaces of non-positive, in particular negative, curvature and in topological dynamics (see [Gr1]). One still does not know how to embrace curvature $K \leqslant 0$ (in particular, 1/5-groups and alike) into a consistent (semi)hyperbolic theory. The obvious difficulty is a non-stability of the condition $K \leqslant 0$ in contrast to the stability of $K < 0$. For example, one does not know how to see semihyperbolicity of a 1/5-group if it is given by a non-1/5-presentation. (Notice that 1/6-groups are caught into our word hyperbolic maze.) But the case $K < 0$ is understood (at least on the foundational level) and an exposition is long overdue. There is a "linguistic" difficulty in discussing hyperbolic groups as one translates clear-cut geometric notions into their "quasi" and "approximate" counterparts. (This difficulty would disappear if one could realize every hyperbolic group by isometries on an appropriate space of negative curvature.) The terminology has undergone a slow selection process in several oral presentations of the subject and is now bearable, I hope. The major clarification was achieved at my stay with Tata Institute in 1984-85 thanks to the receptive audience and many stimulating discussions with G. Prasad. The decision of writing everything up was made under the friendly pressure by S. Gersten at MSRI in Berkeley.

Finally, I wish to thank Jim Cannon who generously spent his time on reading this manuscript, found a multitude of errors and suggested many improvements. He also told me of his work in collaboration with D. Epstein and W. Thurston on an automata theoretic approach to group theory that stems from his paper [Ca]. This approach seems deeper in many respects than the "hyperbolic philosophy" and it applies to a wider class of groups.

<u>The structure of the paper</u>: We start in Sections 1-5 with an

exposition of examples and basic properties of hyperbolic groups. We give few detailed proofs at this stage but rather explain underlying geometric ideas. These ideas become rigorous as we develop in Sections 6-8 geometry of *hyperbolic metric spaces* (see 1.1). In Section 6 we prove the equivalence of different notions of hyperbolicity. This allows us to translate basic facts on manifolds with $K < 0$ to a "curvature free" language which is also "stable" under small perturbations of metrics (compare [Kl]). This stability is achieved at the cost of "quasi"-fication of terminology. There is also a large amount of numerical constants (e.g. constant δ in 0.1) which are needed for definitions of various "quasi"-notions and for establishing relations between them. We tried, whenever possible, to use specific values of the constants and write $A \leqslant 100B$ rather than: "there exists $C \leqslant 100$, such that $A \leqslant CB$."

The discussion in Sections 6 and 7 is purely geometrical. That is no group appears there. Groups Γ came back into play in Section 8, as isometry groups of hyperbolic spaces X; and algebraic properties of Γ are reflected in geometry of X and X/Γ. Our major tool (which does not come free unlike the case $K < 0$) is (a kind of) a *geodesic flow* over X/Γ which completes our dictionary and justifies the treatment of word hyperbolic groups as of fundamental groups of compact manifolds (and spaces) with curvature $K < 0$.

Our exposition is essentially self-contained. Yet a reader may benefit by looking through Chapter V in [Ly-Sc] on small cancellation theory and Chapter I in [B-G-S] on manifolds with curvature $K \leqslant 0$. Further references can be found in these books. Unfortunately, there is a fair amount of "well known" and "easy to prove" little geometric theorems which are hard to locate in the literature. We bring them forth whenever they help to motivate our abstract discussion but we do not provide the proofs.

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§1. Examples and basic properties of hyperbolic groups

1.1 Hyperbolic metric spaces: If X is a metric space then the distance between two points x and y in X is denoted by $|x-y|$. If X is a *pointed space*, that is if some $x_0 \in X$ is chosen as a *reference point*, then we set $|x| = |x|_{x_0} = |x-x_0|$ and

$$(x.y) = (x.y)_{x_0} = 1/2(|x| + |y| - |x-y|).$$

Call X *hyperbolic* with respect to x_0 if it satisfies the following *δ-inequality* (compare 2.4.C.)

$$(*) \qquad (x.y) \geqslant \min((x.z),(y.z)) - \delta$$

for a fixed $\delta \geqslant 0$ and all x, y and z in X. If we want to specify δ we say X is *δ-hyperbolic* with respect to x_0.

Observe the following obvious

1.1.A Lemma: *If X is δ-hyperbolic then*

$$(**) \qquad (t.y) + (z.x) - \min((t.z) + (y.x)\ ,\ (x.t) + (y.z)) \geqslant -2\delta$$

for all x, y, z and t in X.

1.1.B Corollary: *If X is δ-hyperbolic with respect to $x_0 \in X$ then it is 2δ-hyperbolic with respect to every point $x \in X$.*

Proof: By an obvious computation the left-hand side of (**) equals $(t.y)_x - \min((t.z)_x,(y.z)_x)$. Q.E.D.

1.1.C Definition: A metric space X is called *δ-hyperbolic* if it is δ-hyperbolic with respect to each point $x \in X$. We call X *hyperbolic* if it is δ-hyperbolic for some $\delta \geqslant 0$.

1.1.D **Important remark.** Basic examples of hyperbolic spaces are *simplicial trees* (see 1.4.), where the "product" (x.y) equals the distance from the reference point to the edge joining x and y (see 1.4.B.). This shows trees are δ-hyperbolic for $\delta = 0$ (compare 1.4.A.). In fact, every 0-hyperbolic space isometrically embeds into a tree and an arbitrary δ-hyperbolic space can be approximated by trees (see §6).

1.2 First examples of hyperbolic spaces:

(a) The Euclidean space $\mathbb{R}^n$ for $n \geqslant 2$ (obviously) is not hyperbolic but the real line $\mathbb{R}^1$ is δ-hyperbolic for $\delta = 0$.

(b) If X is a bounded space, that is if

$$\operatorname{Diam} X \overset{\text{def}}{=} \sup_{x,y} |x-y| \leqslant d < \infty,$$

then (obviously) X is δ-hyperbolic for $\delta = d$.

(c) Take an arbitrary metric space $(X, |\ |)$ and define a new metric on X by

$$|x-y|' = \log(1 + |x-y|).$$

Then by the *triangle inequality*,

$$|x-y| \leqslant |x-z| + |y-z|,$$

and hence, the new metric satisfies

$$|x-y|' \leqslant \max(|x-z|', |y-z|') + \log 2.$$

It follows (compare 1.1.A) that

$$-|x-y|' - |z-t|' + \max(|x-z|' + |y-t|', |x-t|' + |y-z|')$$

$\geqslant -2 \log 2.$

A straightforward computation identifies the left-hand side of this inequality with

$$(t.y)'_x - \min((t.z)'_x,(y.z)'_x),$$

which shows that $(X,|\ |')$ is δ-hyperbolic for $\delta = 2 \log 2$.

1.3 **Maximal metric:** Consider a non-empty class of metrics on X satisfying a certain condition $\mathcal{C}$. Then the supremum of the metrics in this class is again a metric, unless it becomes infinite at some pair of points in X. This supremum is called the *maximal metric satisfying* $\mathcal{C}$.

1.3.A **Example:** Take a locally finite covering of X by subsets $X_i \subset X$, $i \in I$, where each X_i is endowed with a metric $|\ |_i$. Then we have the maximal metric $|\ |$ on X satisfying $|\ | \leqslant |\ |_i$ on every subset X_i. This metric is nowhere infinite (i.e. an actual metric on X) if and only if the nerve of the covering is connected.

1.4 **Simplical metrics and trees:** A metric $|\ |$ on a simplical polyhedron P is called *simplical* if

(a) the restriction $|\ |_i$ of $|\ |$ on each simplex $\Delta_i \subset P$, $i \in I$, is *Euclidean*. That is $(\Delta_i,|\ |_i)$ admits an affine isometric map into $\mathbb{R}^k$ for $k = \dim \Delta_i$;

(b) the metric $|\ |$ maximal for the condition $|\ | = |\ |_i$ on each Δ_i.

For example, for each $d > 0$ there exists a unique simplical metric on P, where every edge (i.e. a 1-simplex in P) is isometric to $[0,d]$.

1.4.A Recall that P is called *a tree* if it is contractible and dim P = 1. Observe that every two distinct points x and y in a tree P can be joined in P by a unique topological segment, denoted $[x,y] \subset P$. Furthermore, for every finite subset $Y = \{x,y,z,t\ldots\} \subset P$ there is a unique minimal subtree (for some refinement of P) $T(Y) \subset P$ containing Y. Namely, T(Y) is the union of the segments between the points in Y. See Figure 1 below for typical trees T(Y), where Y contains 2, 3 and 4 points.

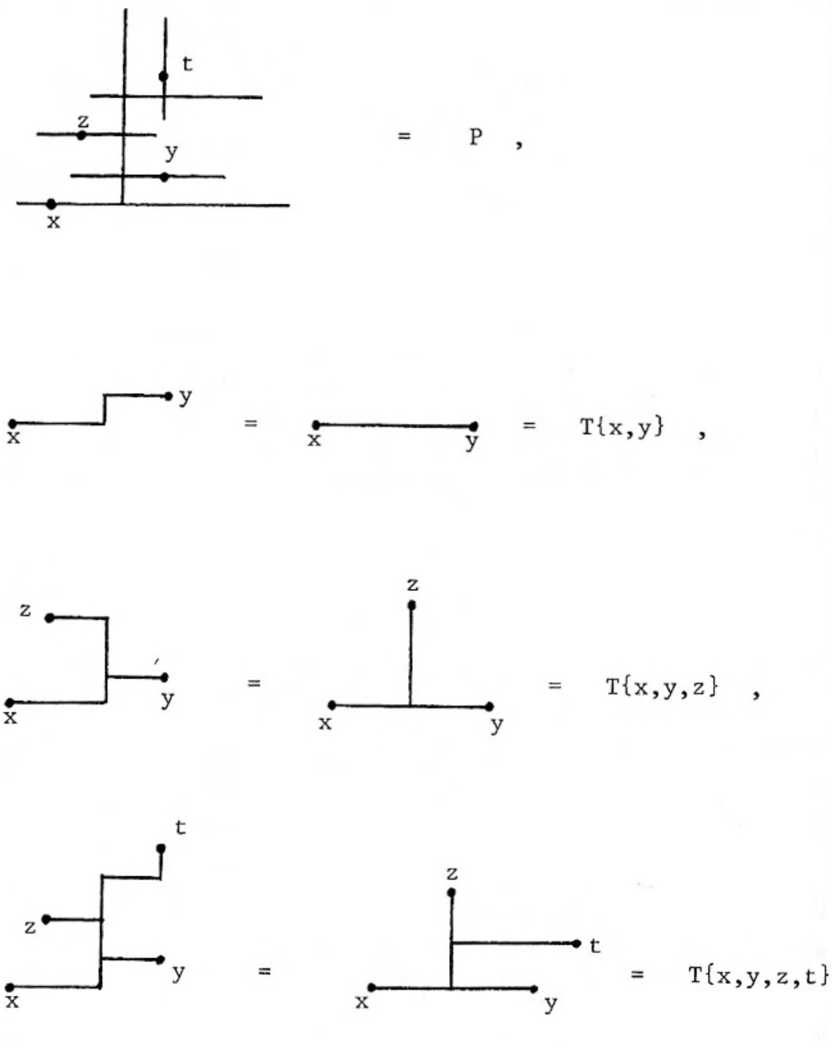

Figure 1

<u>An</u> <u>important</u> <u>example</u>: *Every tree* P *with a simplical* *metric* $|\ |$ *is* δ*-hyperbolic for* $\delta = 0$.

__Proof__: Every segment $[x,y] \subset P$ is divided by the vertices of P meeting $[x,y]$ into finitely many subsegments, say $[p_i,p_j] \subset [x,y]$ isometric to the standard segments $[0,|p_i-p_j|] \subset \mathbb{R}$. Since the metric $|\ |$ is maximal, the total length of these segments equals $|x-y|$. Now, to show that $(t.y)_x \geqslant \min((t.z)_x,(y.z)_x)$, we look at the subtree $T = T\{x,y,z,t\} \subset P$. We ignore the "irrelevant" vertices, where T (and hence P) is locally homeomorphic to $\mathbb{R}$ and thus reduce the story to a tree with at most 6 vertices and 5 edges. We conclude the proof by staring at $T\{x,y,z,t\}$ in Figure 1.

1.4.B __Remark__: In the case of trees the "product" $(y.t)_x$ equals $\mathrm{dist}(x,[y,t])$, where the distance to a subset is measured by

$$\mathrm{dist}(x,Y) \overset{\mathrm{def}}{=} \inf_{y\in Y} |x-y|.$$

1.5 __Geometric examples of hyperbolic spaces__: The following statements (1) and (2) are well known (and quite easy to prove).

(1) Every complete simply connected Riemannian manifold with *strictly negative sectional curvature*,

$$K \leqslant -\varkappa^2 < 0,$$

is δ-hyperbolic for $\delta \leqslant \mathcal{C}\varkappa^{-1}$, where $\mathcal{C}$ is a universal constant in the interval $0 < \mathcal{C} < 10$ (compare 2.4.E.).

(2) Lie groups $O(n,1)$, $U(n,1)$, $Sp(n,1)$ with the left invariant Riemannian metrics and the isometry group of the hyperbolic Cayley plane are hyperbolic. In fact these are the only simple non-compact Lie groups which are hyperbolic for left-invariant *Riemannian* metrics. (One has with example 1.2.(c) an invariant non-Riemannian hyperbolic metric on every Lie group.)

(3) Let $\Omega \subset \mathbb{R}^2$ be a connected and simply connected open subset

which contains no disk of radius $\geqslant R_0$ for some fixed $R_0 > 0$. Define the distance in Ω as the infimum of lengths of curves in Ω between pairs of points. Then Ω with this metric is δ-hyperbolic for some $\delta \leqslant \mathcal{C} R_0^{-1}$ where $0 < \mathcal{C} < 10$. The proof is easy and left to the reader.

Notice that this length metric in Ω can be defined as the maximal metric $|\ |'$ in Ω satisfying $|x-y|' = |x-y|_{\mathbb{R}^2}$ for every pair of points x and y in Ω lying on a straight segment contained in Ω.

1.6 <u>Geodesics metric spaces</u>: A *geodesic segment between two points* x and y in a metric space X is (the image of) an isometric map $[0,|x-y|] \longrightarrow X$ sending $0 \longmapsto x$ and $|x-y| \longmapsto y$. Such a segment, (if it exists at all) may be not unique. Yet we use the notation $[x,y] \subset X$ for such segments. We also denote by $(1-s)x + sy \in [x,y] \subset X$ the image in X of the point $s|x-y| \in [0,|x-y|]$ for $0 \leqslant s \leqslant 1$. If X is given a reference point then sx denotes a point in X such that $|sx| = s|x|$ and $|sx-x| = (1-s)|x|$. If x can be joined by a segment with the reference point, then (obviously) some sx exists for $0 \leqslant s \leqslant 1$.

A metric space X is called *geodesic* if every two points in X can be joined by a segment. It is well known (and easy to prove) that

1.6.A (Hopf-Rinow) *Every complete Riemannian manifold is geodesic.*

1.6.B *Let* P *be a polyhedron with a simplical metric such that every edge has a fixed length* d_0. *Then* P *is geodesic. (The proof of this is an exercise for the reader.)*

1.6.C Let X be a metric space with an arbitrary metric $|\ |$ and denote $|\ |_\epsilon$ for some $\epsilon > 0$ the maximal metric, such that $|x-y|_\epsilon = |x-y|$ for all x and y in X having $|x-y| \leqslant \epsilon$. Note that if $(X,|\ |)$ is a geodesic space then, obviously $|\ |_\epsilon = |\ |$ for all $\epsilon > 0$. Let $|\ |_0 = \sup_{\epsilon>0} |\ |_\epsilon$ and assume $|\ |_0$ is a (nowhere infinite) metric in X. Then by an easy argument the metric completion of $(X,|\ |_0)$ is a geodesic space.

Example: Let X be a smooth connected submanifold in $\mathbb{R}^n$ with the induced metric. Then $|\ |_0$ equals the *induced Riemannian metric* in X, where the distance between two points equals the infimum of lengths of smooth curves in X between these points.

1.7 Polyhedron $P_d(X)$: Let X be a metric space, take $d \geqslant 0$ and let $P_d(X)$ be the simplical polyhedron whose set of vertices (i.e. the 0-skeleton) equals X and where a finite subset $Y \subset X$ spans a simplex in $P_d(X)$ if and only if the distance between every two points in Y is $\leqslant d$. Notice that if X is a geodesic space then the 1-skeleton of $P_d(X)$ is equal to that of the nerve of the covering of X by the balls of radius $\leqslant 1/2d$. Recall that the ball in X of radius R around $x \in X$ is

$$B_x(R) = \{y \in X \mid |x-y| \leqslant R\}.$$

Remark: The space $P_d(X)$ serves as a regularization of X. For example, $P_d(X)$ allows us to state Is_2 (see §0) for an arbitrary X and then to prove Is_2 for hyperbolic spaces X (see 1.7.C.). One also needs $P_d(X)$ in the proof of Rips' theorem on the finiteness of the virtual homological dimension of word hyperbolic groups (see 1.7.A., 1.7.D. and 2.2.).

Let K be a subpolyhedron in $P_d(X)$ and let $Y \subset X$ denote the set of vertices of K. Observe that every map $f\colon Y \longrightarrow X$ defines a simplical map, say $f'\colon K \longrightarrow P_{d'}(X)$, provided $d' \geqslant |f(y_1)-f(y_2)|$ for all pairs y_1 and y_2 in Y satisfying $|y_1-y_2| \leqslant d$. Furthermore, if $d' \geqslant d$ and $|y_1-f(y_2)| \leqslant d'$ for $|y_1-y_2| \leqslant d$, then (obviously) there exists a homotopy in $P_{d'}(X)$ between map f' and the original embedding $K \subset P_d(X) \subset P_{d'}(X)$.

The following important observation is due (up to the terminology) to I. Rips.

1.7.A **Lemma:** *Let* X *be a* δ*-hyperbolic space such that every* $x \in X$ *can be joined by a segment with a fixed reference point* x_0 *in* X. *Then the polyhedron* $P_d(X)$ *is contractible for all* $d \geqslant 4\delta$.

Proof: It suffices to contract every finite subpolyhedron K in $P_d(X)$ which is done by moving the vertices $y \in X$ of K one after another toward the reference point x_0 in X. To see how it works take a vertex, say $y_0 \in X$ of K, move it to $y_0' = |y_0|^{-1}(|y_0|-d_0)y_0$ in X for some $d_0 > 0$ and estimate $|x-y_0'|$ for all $x \in X$ as follows

$$\begin{aligned}|x-y_0'| &= |x| + |y_0'| - 2(x.y_0') \leqslant \\ &\leqslant 2\delta + |x| + |y_0'| - 2\min((x.y_0),(y_0.y_0')) = \\ &= 2\delta + \max(|x-y_0|-d_0, |x|-|y_0|+d_0).\end{aligned}$$

If y_0 is the farthest vertex of K from x_0, then this inequality implies for all vertices y of K and for $2\delta \leqslant d_0 \leqslant 1/2d$, $|y-y_0| \leqslant 2\delta + \max(d-d_0,d_0) \leqslant d$. Hence, this move defines a homotopy of K in $P_d(X)$ and finitely many such homotopies obviously contract K in $P_d(X)$.

1.7.B **Remark:** Suppose K is homeomorphic to the circle and let y_1 and y_2 be the vertices in K adjacent to the farthest (from x_0) vertex y_0. Then we see as earlier,

$$\begin{aligned}|y_1-y_2| &\leqslant 2\delta + \max(|y_1-y_0|+|y_2|-|y_0|, |y_2-y_0|+|y_1|-|y_0|) \\ &\leqslant 2\delta + d.\end{aligned}$$

Consider two kinds of homotopy of K to a shorter circle K' in $P_d(X)$.

Case 1: If $|y_1-y_2| \leqslant d$ then the shortening of K is obvious: remove the edges $[y_1,y_0]$ and $[y_2,y_0]$ and join y_1 with y_2 by an edge in

$K_d(X)$.

__Case 2__: Let $|y_1-y_2| \geqslant d$. Then by the above inequality one of the vertices y_1 and y_2, say y_1, satisfies

$$|y_1| \geqslant |y_0| - 2\delta.$$

Now, move $y_1 \longmapsto y_1' = |y_1|^{-1}(|y_1|-d_1)y_1$ and observe that

$$|y-y_1| \leqslant 2\delta + \max(d-d_1, d_1+2\delta)$$

for all vertices $y \in X$ of K and

$$|y_1-y_2| \leqslant 2\delta + \max(d+2\delta-d_1, d_1+2\delta).$$

Therefore, if $4\delta \leqslant d_1 \leqslant d - 4\delta$, then the move $y_1 \longmapsto y_1'$ yields a homotopy of K in $P_d(X)$ bringing the vertices y_1 and y_2 within distance $\leqslant d$ in X, which allows one to apply the Case 1 shortening of K.

Notice that the shortening homotopy in Case 2 "consists" of three triangles: two take care of the move $y_1 \longmapsto y_2'$ and the third comes from the Case 1 homotopy. So the total number of triangles needed to contract K is $3N_1$ where N_1 is the number of edges in K. In fact, what we have just proved can be expressed more precisely in the following

1.7.C __Lemma__: *Let the circle* S^2 *be subdivided into* N_1 *segments and let* $f\colon S^1 \longrightarrow P_d(X)$ *be a simplical map. If* $d \geqslant 8\delta$, *then there exist a triangulation of the disk* D^2 *into* $N_2 \leqslant 3N_1$ *triangles and a simplical map* $D^2 \longrightarrow P_d(X)$ *which extends* f *from the boundary* $\partial D^2 = S^1$.

1.7.D __Remark:__ The proof of 1.7.A also yields the following generalization of 1.7.A.

Let X *be a* δ*-hyperbolic space which satisfies the following condition* (*) *for some* $\epsilon > 0$ *and* $d > 0$,

(*) *for every* $x \in X$ *with* $|x| \geqslant (1/2)d$ *there exists a point* $x' \in X$, *such that* $|x'| \leqslant \min(|x|-\epsilon, |x|-2\delta)$ *and* $|x-x'| \leqslant d - 2\delta$.

Then the polyhedron $P_d(X)$ *is contractible.*

1.8 **Hyperbolic boundary ∂X of X:** A sequence $x_i \in X$, $i = 1,2,\ldots$, is called *convergent at infinity* if

$$(x_i.x_j) \longrightarrow \infty \text{ for } i,j \longrightarrow \infty.$$

Observe that this convergence is independent of the choice of the implied reference point in X. Furthermore, if X is hyperbolic then the equality

$$\lim_{i,j\to\infty} \inf(x_i.y_j) = \infty$$

is an equivalence relation on the set of sequences in X convergent at infinity.

Definition: The (hyperbolic) *boundary* ∂X of a hyperbolic space is the set of the equivalence classes of sequences in X convergent at infinity. If a sequence x_i is contained in the class $a \in \partial X$ we write $x_i \longrightarrow a$ for $i \longrightarrow \infty$. Thus we define a natural topology on $X \cup \partial X$, such that X is dense in $X \cup \partial X$. Observe that this topology is independent of the choice of the reference point in X. It follows that the isometry group of X acts on ∂X by *homeomorphisms* of ∂X. More generally, every isometric map between hyperbolic spaces, say $f\colon X \longrightarrow Y$, extends to a unique topological embedding $X \cup \partial X \longrightarrow Y \cup \partial Y$ sending ∂X to ∂Y. Recall that a map f is *isometric* if $|f(x_1)-f(x_2)| = |x_1-x_2|$ for all x_1 and x_2 in X.

1.8.A **Examples:**

(a) Let X be a tree with a simplical metric. Then the boundary ∂X is canonically (and obviously) homeomorphic to the space of isometric maps $f\colon \mathbb{R}_+ \longrightarrow X$ sending zero to the reference point in X, where the space of maps is given the point-wise convergence topology. It easily follows that the topological dimension of this boundary equals zero. If the tree X is locally finite, then the boundary ∂X (obviously) is compact.

(b) Let X_0 be an arbitrary metric space and let $f(t)$ for $t \in (-\infty,+\infty)$ be a positive monotone increasing function such that $f(t+1) \geqslant \lambda f(t)$ for all $t \in \mathbb{R}$ and some fixed $\lambda > 1$. Consider the maximal metric $|\ |$ on $X = X_0 \times \mathbb{R}$ for which the embedding $\mathbb{R} \longrightarrow X$ given by $t \longmapsto (x_0,t)$ is isometric for all $x_0 \in X_0$ and

$$|(x_1,t)-(x_2,t)| \leqslant f(t)|x_1-x_2|_{X_0}$$

for all $t \in \mathbb{R}$ and $x_1,x_2 \in X_0$.

One can easily see that the space $(X,|\ |)$ is hyperbolic and the correspondence $x_0 \longmapsto (x_0,t_i)$ for $x_0 \in X_0$ and $t_i \longrightarrow +\infty$ defines an embedding of X_0 into an open subset in ∂X whose complement is a single point, say $a_0 \in \partial X$, such that $(x_0,t_j) \longrightarrow a_0$ for every fixed $x_0 \in X_0$ and for $t_j \longrightarrow -\infty$. Notice that for $X_0 = \mathbb{R}^n$ and $f(t) = e^t$ the space X is the ordinary hyperbolic space H^{n+1} and $\partial H^{n+1} = S^n = \mathbb{R}^n \cup \{\infty\}$.

(b′) Let us replace $\mathbb{R}$ by $\mathbb{R}_+$ in the above construction. The resulting space $X \approx X_0 \times \mathbb{R}_+$ is not necessarily hyperbolic. In fact, it is hyperbolic if and only if X_0 is hyperbolic, as a simple argument shows. The boundary of such a hyperbolic X is canonically homeomorphic to X_0.

1.8.B Let us show that *the boundary ∂X of an arbitrary hyperbolic space X is metrizable.* In fact, let us define $|a-b| = |a-b|_{x_0}$ for $a,b \in \partial X$ to be the maximal metric satisfying the following condition:

if $x_i \longrightarrow a$ and $x_j \longrightarrow b$, and $\liminf_{i,j\to\infty} (x_i.x_j) \geq 2^k$ for some $k = 1,2,\ldots$ then $|a-b| \leq k^{-1}$.

A simple argument (left to the reader) shows that $|a-b|$ is indeed a metric giving the right topology to ∂X and that a change of the reference point $x_0 \in X$ leads to an *equivalent* metric on ∂X. That is the identity map

$$(\partial X, |\ |_{x_0}) \longleftrightarrow (\partial X, |\ |_{x_1})$$

is a *uniform* (i.e. uniformly continuous in both directions) homeomorphism for all x_0 and x_1 in X. Also notice that our metric on ∂X is *complete*. The proof of this is left to the reader.

<u>Remark</u>: If X is a complete Riemannian manifold or, more generally a geodesic space where all closed balls are compact, then the boundary ∂X is compact. Furthermore, if such an X is unbounded then ∂X is non-empty.

<u>Counterexample</u>: Let X be the union of the segments [0,i], for i = 1,2,..., joined at zero. This X is unbounded geodesic hyperbolic but ∂X is empty.

§2. Hyperbolic metric groups

An abstract group Γ with a left invariant metric $|\ |$ on Γ is called a *metric group*. We always consider Γ as a pointed space where the preferred point is the identity element in Γ.

2.1 Word metrics: Let a subset $G \subset \Gamma$ generate Γ. The corresponding *word metric* $|\ |_G$ is the maximal metric on Γ satisfying $|g| = |g^{-1}| = 1$ for all $g \in G$. If Γ is finitely generated, then the expression "a word metric on Γ" always refers to a word metric for some *finite* generating subset G in Γ.

Observe that our definition agrees with the ordinary definition as $|\gamma_1 - \gamma_2|_G$ equals the length of the shortest word in g and g^{-1} for $g \in G$ representing $\gamma_2^{-1}\gamma_1 \in \Gamma$.

2.1.A Definition: A metric group Γ is called *hyperbolic* (δ-hyperbolic) if it is hyperbolic (δ-hyperbolic) as a metric space.

A finitely generated group Γ is called *word hyperbolic* if some word metric on Γ is hyperbolic.

We shall see later that the word hyperbolicity is independent of the choice of a finite generating set $G \subset \Gamma$ but the implied constant δ may depend on G.

2.2 Contractability of $P_d(\Gamma)$.

Theorem (I. Rips): *Let Γ be a word hyperbolic group. Then Γ admits a faithful discrete simplical action on some finite dimensional locally compact and contractible polyhedron* P, *such that* P/Γ *is compact.*

Proof: Take $P = P_d(\Gamma)$ (see 1.7) with the obvious action of Γ on P. Since Γ is finitely generated, P obviously satisfies all requirements with the possible exception of the contractibility.

Now, if Γ is δ-hyperbolic, then it satisfies (*) in 1.7.D for $\epsilon = 1$ and all $d \geqslant 4\delta + 1$ since the metric in question is a *word* metric. Hence, $P_d(\Gamma)$ is contractible for $d \geqslant 4\delta + 1$. Q.E.D.

<u>Corollaries</u>:

2.2.A *The group Γ is finitely presented.*

2.2.B *The group Γ contains at most finitely many conjugacy classes of torsion elements.*

2.2.C *The rational cohomology $H^*(\Gamma,\mathbb{Q})$ is finite dimensional. Furthermore, the L_2-cohomology (see* [C-G]) *of Γ is Γ-finite dimensional.*

2.3 <u>Isoperimetric inequalities for hyperbolic groups</u>: Let Γ be *presented* by $\{G \mid W\}$, where $G \subset \Gamma$ is a generating set and W is a subset in the free group $\mathcal{F}(G)$ freely generated by G, such that W lies in the kernel $\mathcal{N} \subset \mathcal{F}(G)$ of the obvious epimorphism $\mathcal{F}(G) \longrightarrow \Gamma$ and W *normally generates* $\mathcal{N}$, that is $\mathcal{N} = \mathcal{N}(W)$ is the *minimal* normal subgroup in $\mathcal{F}(G)$ containing W.

Recall that every $f \in \mathcal{F}(G)$ is an irreducible word $g_1^{a_1} g_2^{a_2} \dots g_k^{a_k}$ for $g_i \in G$ and $a_i \in \mathbb{Z}$ and define the *length* $L(f)$ of f by

$$L(f) = |a_1| + |a_2| + \dots + |a_k|.$$

Next define the *area* $A(f)$ for all $f \in \mathcal{N}(W)$ as the minimal number A, such that f can be written in the form

$$f = \rho_1 w_1^{b_1} \rho_1^{-1} \rho_2 w_2^{b_2} \rho_2^{-1} \dots \rho_m w_m^{b_m} \rho_m^{-1}$$

for some $\rho_i \in \mathcal{F}(G)$ and $w_i \in W$, $i = 1,\dots,m$, where

$$L(\rho_1) + L(\rho_2) + \dots + L(\rho_m) \leqslant A$$

and

$$|b_1| L^2(w_1) + |b_2| L^2(w_2) + \dots + |b_m| L^2(w_m) = A.$$

2.3.A **Theorem:** *Let* Γ *be* δ*-hyperbolic for (the word metric associated with) a generating subset* $G \subset \Gamma$ *and let* $W = W_d \subset \mathcal{F}(G)$, *for some* $d \geq 0$, *be the set of those irreducible words (in letters* $g \in G$) *of length* $\leq 3d$ *which are equal to the identity element in* Γ. *If* $d \geq 8(\delta+2)$, *then* $\Gamma = \mathcal{F}(G)/\mathcal{N}(W)$ *and*

$$A(f) \leq 27\ d^2 L(f)$$

for all $f \in \mathcal{N}(W)$.

Proof: If Γ has no torsion of order ≤ 3, then the (simply connected!) complex $P_d^2(\Gamma)$ is acted on *freely* by Γ and the theorem follows from 1.7.A and 1.7.C since every triangle in $P_d^2(\Gamma)$ defines a word in W_d. Notice that the inequality $d \geq 8\delta$ suffices in this case.

Now, in the general case we recall the Cayley complex (graph) X of (Γ, G) (if $G \cap G^{-1} = \emptyset$ then $X = P_1^1(\Gamma)$) and then apply 1.7.C to some simply connected subcomplex P in $P_d^2(X)$ in which the action of Γ is free. We leave to the reader the (obvious) details of the argument.

2.3.B **Corollary:** *If* Γ *is word hyperbolic (for some finite generating subset in* Γ) *then for every finite presentation* $\{G | W\}$ *of* Γ *there exists a constant* $C \geq 0$, *such that*

$$A(f) \leq CL(f)$$

for all $f \in \mathcal{N}(W)$. *In particular, the word problem is solvable in* Γ.

2.3.C **Remark:** We shall see in 7.4 that the conjugacy problem also is solvable in Γ.

2.3.D Let us state a converse to 2.3.B (see 6.8 for the proof).

Theorem: *Let* Γ *be presented by* $\langle G|W\rangle$, *such that all* $f \in \mathcal{N}(W)$ *satisfy*

$$A(f) \leqslant CL(f)$$

for some constant $C \geqslant 0$ *(depending on* $\langle G|W\rangle$ *but not on* f). *Then* Γ *is hyperbolic for the word metric associated to* G.

This theorem immediately implies with 2.3.B the following

2.3.E **Corollary:** *If* Γ *is hyperbolic for some finite generating subset* $G_0 \subset \Gamma$ *then* Γ *is hyperbolic for every finite generating subset* $G \subset \Gamma$.

2.3.F Let us give a quantitative version of 2.3.D for *triangular* presentations $\langle G|W\rangle$ where every word $w \in W$ has length $\leqslant 3$. Observe that every presentation can be "triangulated" in an obvious way, which allows us, in principle, to apply the following theorem to non-triangular presentations.

Let $\langle G|W\rangle$ *be triangular and suppose there exists a number* $N \geqslant N_0 = 10000$, *such that every* $f \in \mathcal{N}(W)$ *with the area in the interval*

$$N \leqslant A(f) \leqslant 100N$$

satisfies

$$L(f) \geqslant 1000\sqrt{A(f)}.$$

Then the group Γ *with the word metric associated to* G *is* δ*-hyperbolic for* $\delta = 1000\sqrt{N}$. *Furthermore all*